Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: LegisiMate
b. Team leader name: Vachaspati Goutam
c. Problem Statement: Accessible, evidence-backed legal case preparation (Social Good)

Brief about the idea

LegisiMate turns a person's case facts into an evidence-backed preparation file before they meet an advocate.

- **Problem:** in Indian cheque-dishonour cases (Section 138, NI Act), strict statutory clocks, missing proof of notice and confused dates decide cases long before any argument. Generic chatbots guess dates and invent citations.
- **Solution:** deadlines are checked in code, Gemini on Vertex AI researches a local corpus of Indian law, three independent mock hearings test both sides, and a readiness brief lists what to collect and what to ask.
- **For:** individuals, small businesses (MSMEs), junior advocates, law students and legal-aid clinics.
- **Boundaries:** preparation support, not legal advice. No verdict prediction; missing facts stay visibly missing.

7,143
statutory provisions indexed

1,600
judgments / 10,139 searchable chunks

3 x 22
independent hearing paths x statements

Proof of concept: one Indian case type. Vision: constitutions, legislation and case law for every country with a constitution.

Opportunities

- **How different is it from any of the other existing ideas?**
  Legal chatbots answer from model memory, guess dates and can invent citations. LegisiMate splits the work: code computes every statutory deadline, the model may only quote passages it retrieved, and the server attaches and verifies the exact source text. Unknown facts are never filled in.
- **How will it be able to solve the problem?**
  A guided intake separates confirmed from unconfirmed dates; a bounded research agent finds statutes and precedents; three fresh-memory hearings test the case from both sides; the brief lists missing evidence, next steps and questions for an advocate. Users arrive at a consultation with a better file.
- **USP of the proposed solution**
  Source-verified, adversarially rehearsed preparation instead of a verdict. Every quote is traceable, every date rule is code, and a 'chance of winning' is deliberately absent.

List of features offered by the solution

Guided intake and tour
Six-step tour on a fictional sample. Consent required before any AI call.

Deterministic calendar review
30-day notice, 15-day payment window, one calendar month to file. Missing dates stay unresolved.

Agentic legal research
Gemini chooses statute and precedent searches, source reads and deadline checks (max 4 steps).

Local legal RAG
SQLite FTS5 index of 7,143 provisions and 10,139 chunks from the SC, Delhi HC and Bombay HC.

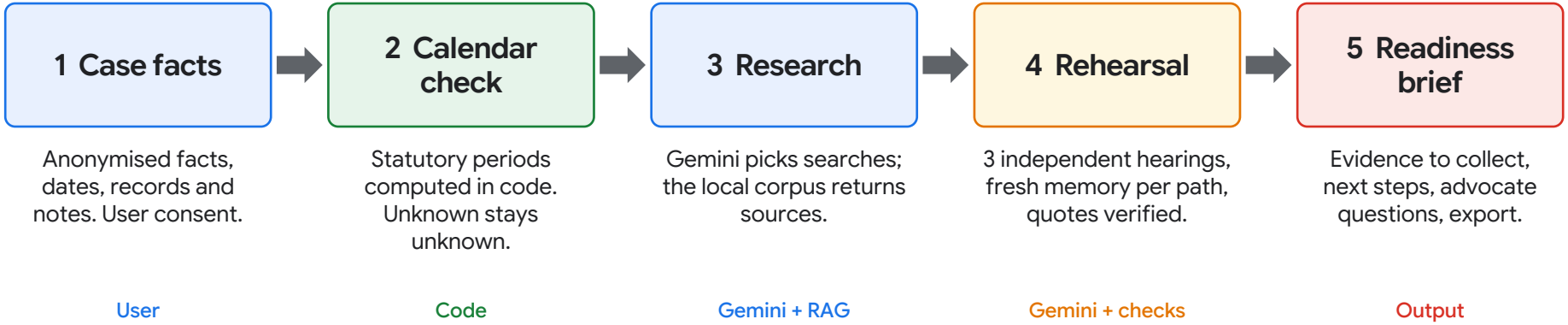
Three mock hearings
Magistrate, counsel, parties, witnesses and juniors: openings, cross-examination, rebuttals, assessment.

Source-bound quotations
Model cites passage IDs; the server attaches the exact corpus text and checks the match.

Decision tree and courtroom replay
Explore each path by hearing stage; 2.5D replay with play, pause, speed and reduced motion.

Readiness brief and export
Evidence to collect, next steps, questions for an advocate, settlement options; Markdown export.

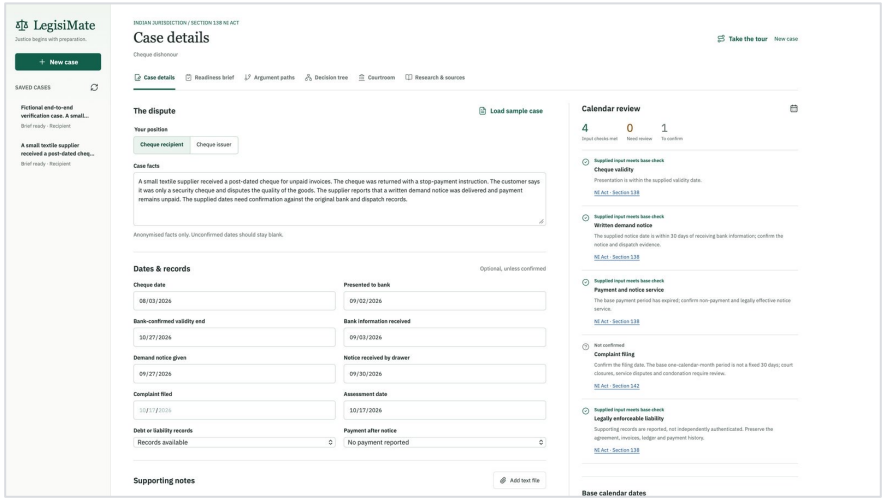
Process flow diagram or Use-case diagram



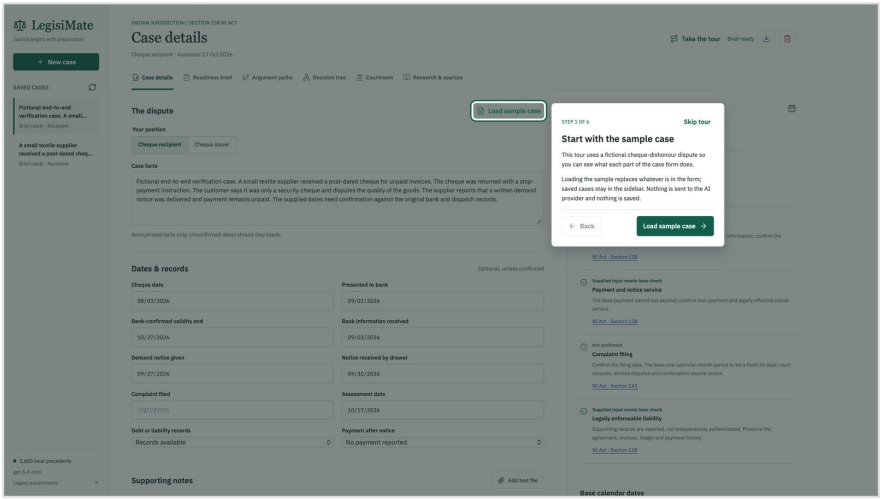
Use case

A small textile supplier holds a dishonoured cheque for unpaid invoices. Before paying for a consultation, they check whether the demand-notice and complaint dates still work, see which proof of service is missing, rehearse what the drawer's counsel will argue, and leave with a checklist and questions for their advocate.

Wireframes/Mock diagrams of the proposed solution (optional)



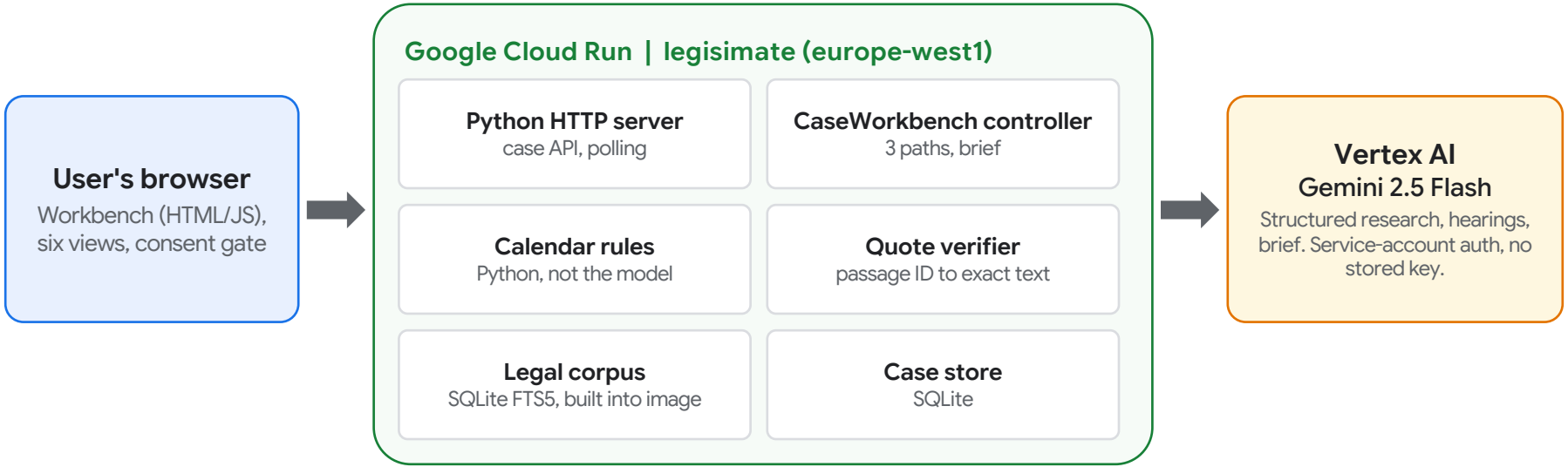
Case details: dates, records and live calendar review



Guided tour over the real form, using a fictional sample

Six views in one workbench: Case details, Readiness brief, Argument paths, Decision tree, Courtroom, Research & sources. Implemented screens, not mock-ups.

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google Cloud

- Vertex AI: Gemini 2.5 Flash with structured JSON output
- Cloud Run: serverless container, scales to zero
- Cloud Build: CI/CD trigger on GitHub master
- Artifact Registry: container images
- Also supports a Google AI Studio Gemini API key

Backend and AI

- Python 3.12, standard-library HTTP server
- Pydantic schemas for model output
- SQLite FTS5 (BM25) legal retrieval
- PyArrow: Parquet slice of vaquill/open-india-law
- Deterministic statutory calendar engine

Frontend and delivery

- HTML, CSS and vanilla JavaScript
- Cytoscape.js decision tree
- Lucide icons, IBM Plex Sans
- 2.5D courtroom replay renderer
- Docker image (python:3.12-slim)

Estimated implementation cost (optional)

Hosting: Cloud Run

Request-based billing with 0 minimum instances, so nothing runs while idle. 1 vCPU / 512 MiB per instance. The legal index ships inside the image, so there is no database to pay for.

Build and storage

Cloud Build runs only on a push to master (about 2.5 minutes per build). One container image kept in Artifact Registry.

AI usage: Gemini 2.5 Flash

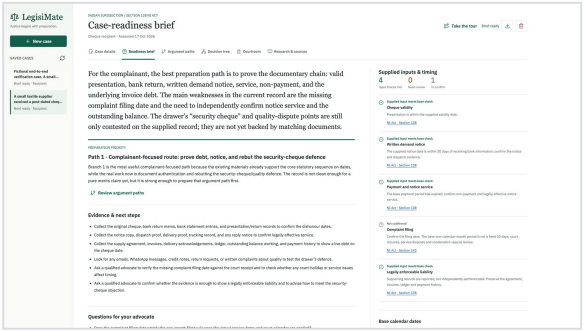
The main variable cost, billed per token at Vertex AI list prices. Each analysis has a hard ceiling: 72 model calls by default, 93 at maximum depth.

Cost controls

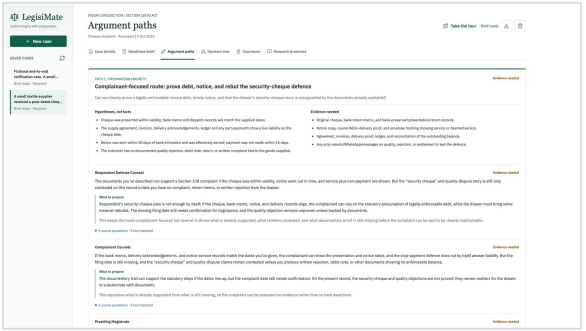
Consent gate before any model call; one analysis at a time; the call budget is shown before submission; showcase cases replay with zero model calls.

Per-analysis token spend has not been measured yet; it is the next cost benchmark. No fixed infrastructure cost while idle.

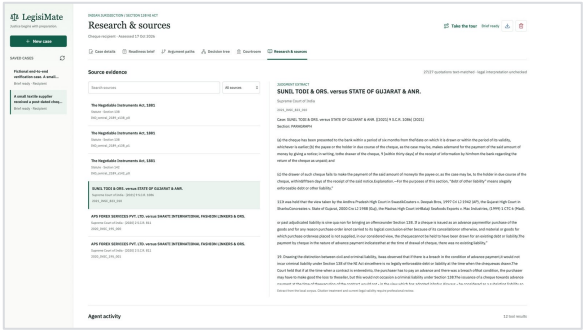
Snapshots of the prototype



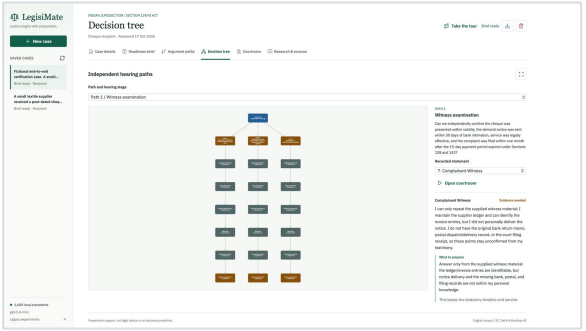
Readiness brief



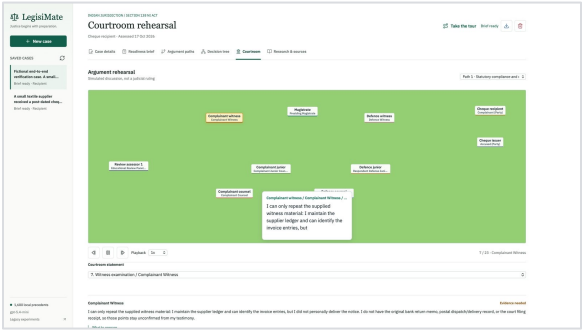
Argument paths



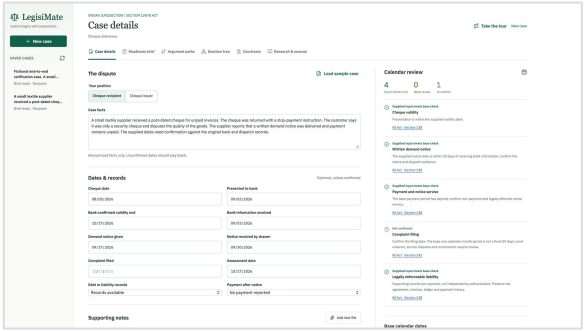
Research and sources



Decision tree



Courtroom rehearsal



Case details and calendar

Prototype Performance report/Benchmarking

Check	Result
Automated tests	124 / 124 backend tests passed, plus Node graph and playback checks
Full live hearing run	69 statements, 10 participants, 73 of 75 allowed calls, 206 / 206 quotes matched source text
Gemini 2.5 Flash showcase	27 / 27 quotes verified across 9 statements and 3 independent paths
Corpus-grounded fallback	66 actions end to end, 132 / 132 quotes matched source text
Legal index build	About 1.5 s from 11.8 MB of Parquet; built into the container image
Deploy pipeline	GitHub push to live Cloud Run revision in about 2.5 minutes
UI stress and layout	12 participants / 87 statements; checked at 390 px mobile and 1440 px desktop

Integration and provenance checks. Matching a quote to its source does not prove the legal reasoning is right; legal accuracy needs qualified evaluation.

Additional Details/Future Development (if any)

Next

- Read PDFs and photos of return memos and notices (OCR)
- Qualified legal evaluation and user studies
- Measure Gemini cost and latency per analysis
- Evaluate hybrid (embedding + BM25) retrieval
- Secure multi-user accounts and retention controls
- Indian languages

Vision

- Constitutions, legislation, regulations and case law for every country with a constitution
- Versioned jurisdiction packs from authoritative, rights-cleared sources (Akoma Ntoso, ELI, USLM)
- Country-specific rules and local evaluation before each release
- Same engine: code for deadlines, Gemini for reasoning, verified sources
- Never a promised win: preparation, human judgement, inspectable evidence

Provide links to your:

- GitHub Public Repository: https://github.com/venunayira/LegisiMate
- Demo Video Link (3 Minutes): https://youtu.be/-pd5MmUemtE
- Final Product Link: https://legisimate-587527883402.europe-west1.run.app/

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Thank you

